Schur's lemma, $B = \mathrm{End}_A(E)$ is naturally identified with $\bigoplus_i \mathrm{End}(W_i)$. This implies all the statements of the theorem. $\square$

We will now apply Theorem 5.18.1 to the following situation: $E = V^{\otimes n}$, where V is a finite dimensional vector space over a field k of characteristic zero and A is the image of $k[S_n]$ in $\mathrm{End}\, E$. Let us now characterize the algebra B. Let $\mathfrak{gl}(V)$ be $\mathrm{End}\, V$ regarded as a Lie algebra with operation $ab - ba$.

Theorem 5.18.2. *The algebra $B = \mathrm{End}_A E$ is the image of the universal enveloping algebra $\mathcal{U}(\mathfrak{gl}(V))$ under its natural action on E. In other words, B is generated by elements of the form*

$$\Delta_n(b) := b \otimes 1 \otimes \cdots \otimes 1 + 1 \otimes b \otimes \cdots \otimes 1 + \cdots + 1 \otimes 1 \otimes \cdots \otimes b,$$

$b \in \mathfrak{gl}(V)$.

Proof. Clearly, the image of $\mathcal{U}(\mathfrak{gl}(V))$ is contained in B, so we just need to show that any element of B is contained in the image of $\mathcal{U}(\mathfrak{gl}(V))$. By definition, $B = S^n \mathrm{End}\, V$, so the result follows from part (ii) of the following lemma.

Lemma 5.18.3. *Let k be a field of characteristic zero.*

(i) For any finite dimensional vector space U over k, the space $S^n U$ is spanned by elements of the form $u \otimes \cdots \otimes u$, $u \in U$.

(ii) For any algebra A over k, the algebra $S^n A$ is generated by elements $\Delta_n(a)$, $a \in A$.

Proof. (i) The space $S^n U$ is an irreducible representation of $GL(U)$ (Problem 4.12.3). The subspace spanned by $u \otimes \cdots \otimes u$ is a nonzero subrepresentation, so it must be everything.

(ii) By the fundamental theorem on symmetric functions, there exists a polynomial P with rational coefficients such that

$$P(H_1(x), \ldots, H_n(x)) = x_1 \ldots x_n$$

(where $x = (x_1, \ldots, x_n)$). Then

$$P(\Delta_n(a), \Delta_n(a^2), \ldots, \Delta_n(a^n)) = a \otimes \cdots \otimes a.$$

The rest follows from (i). $\square$